Foundations of Full House Methodology

Daniel J. Eck[1], Shen Yan[1*]

1. Department of Statistics, University of Illinois Urbana-Champaign

Abstract

The Full House Methodology is an approach for computing era-adjusted or system-adjusted statistics that balances the size of the eligible population in each system with the component's performance via "versus their peers" in a specific system. We assume that each component has a latent aptitude that can be calculated from this balancing act provided the system inclusion mechanism for components of the system is known. Without making any assumptions about the distribution of the system's components, the components' distribution is well-established from the nonparametric probability distribution.

1 Introduction

Several academics have attempted to develop methods that can correct for era or system biases. The starting place for the answer to the problem of comparing players across eras can be found in the work of Stephen Jay Gould. Gould [1996] suggested that the entire distribution of achievement, or "full house of variation," is relevant when considering the achievements of baseball players across eras. This idea was also recognized as the starting place for an answer by Schell [2013] and Schell [2016].

Berry et al. [1999], Petersen et al. [2011], Schell [2013, 2016], Petersen and Penner [2020] provide several techniques to compare baseball players across eras, but they have fallen short in their objective and their results remain biased, as discussed in Eck [2020a] and Eck [2020b]. In short, these previous attempts do not incorporate the evolving talent pool in calculations, which is necessary for understanding the changing distributions of achievement over time. The result is that existing methodologies produce a disproportionate number of players who began their career before baseball was integrated players in the rankings of baseball's greatest of all time. This era of baseball did not include a crucial population of eligible players. Yu et al. [2016], Meyer [2021], Campbell [1990] also develop methods for comparing components in historical popularity, educational attainment, and evaluation of employees. But they have similar problems as stated before and fail to account for the evolving talent pool.

In some circumstances, such as test equating, it is not required to take into account an evolving talent pool. Andersson et al. [2013] offers a collaborative framework for test equating and builds the cumulative density function using the kernel method. Nevertheless,

*Daniel J. Eck (dje13@illinois.edu); email addresses: Shen Yan (shenyan3@illinois.edu)

they are unable to resolve the extrapolation issue in non-parametric estimation, and the distribution's tail is not given special consideration.

Our Full House methodology directly incorporates the quality of the talent pool as a central modeling component. It offers metrics for assessing the performance of components in various eras or systems. With the application of educational attainment simulation, our Full House methodology reveals that the most talented students could perform worse than the other students who attend schools with lower mean scores.

The next section establishes the model setting and specifies the assumptions in our Full House methodology. Section 3 considers an intriguing paradox that the component with more talent underperforms the component with less talent. We compute the probability of this paradox under different scenarios. Section 4 discusses the application of the Full House methodology for the educational attainment simulation.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j. We then define an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$ as a determined and known system inclusion function for population i where a 1 indicates inclusion and a 0 indicates exclusion. We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, $n_i \leq N_i$. As we will see later, the observed $Y_{i,j}$ values are one-to-one realizations of $X_{i,j}$, provided that $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1$. Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. We now form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$ where $r_{i,j}$ indexes the aptitude of the jth individual included in the system. As an example, let $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, then the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. Full House Modeling states this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}}\left(Y_{i,(j)}\right), \end{aligned} \tag{1}$$

where $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarise Full House Modeling in the definition below.

Definition 2.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$ Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$ for all individuals such that $1_i(X_{i,(r_{i,j})}, \boldsymbol{X}_{i,-(r_{i,j})}) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (1) where F_{Y_i} and F_X are continuous and invertible.*

Note that Definition 2.1 fixes F_X but allows F_{Y_i} to vary across systems.

Remarks:

1. The definition of a Full House Model immediately implies structural missingness. Every $X_{i,j}$ such that $1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ will not have a corresponding Y value. In the following sections we will consider probabilities involving both X and Y quantities. To handle this structural missingness, we will suppose that $Y_{i,j}$ values that are not observed are not P-measurable.

3 Paradoxes Under Same Distribution

In this section we will provide explicit paradox probability results in a Full House Modeling setting involving two populations with the additional specification that $F_{Y_i} = F_Y$ in Definition 2.1 for $i = 1, 2$. We define a paradox to have occurred when

$$X_{1,(r_{1,j})} > X_{2,(r_{2,k})} \quad \text{and} \quad Y_{1,(j)} < Y_{2,(k)},$$

where $i = 1, 2$, $1 \leq j \leq n_1$, and $1 \leq k \leq n_2$. Our central interest will be the determination of

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{2}$$

As a simple illustrative example of some mechanics concerning paradoxes, let $r_{i,t} = N_i - n_i + t$ and set $k = n_1, j = n_2$, and further suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then (2) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ as $N_1 \to \infty$. We now introduce a convenient representation which allows the paradox (2) to be expressed in terms of the latent abilities alone.

Lemma 3.1. *Consider a Full House Model in Definition 2.1 with two populations and $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right), \tag{3}$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$.

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{1,2,j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. □

Definition 3.2. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function, the cumulative distribution function of a Beta(α, β) distribution, is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta coefficient.

Lemma 3.1 and Definition 3.2 give enough structure for us to state an prove an probability for a paradox assuming a Full House Model.

Theorem 3.3. *Consider a Full House Model in Definition 2.1 with two populations and $F_{Y_i} = F_Y$ for $i = 1, 2$. Let I be a regularized incomplete beta function. Assume that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \geq 0, \qquad \text{for all } t \in (0,1). \tag{4}$$

Then

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2-n_1-p+k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2-N_1-p+r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The result of Theorem 3.3 is distribution free, and only depends on the sample sizes n_i, N_i, $i = 1, 2$, the system inclusion function, and the indexes of the order statistics under comparison. The distribution on Y is eliminated by Lemma 3.1. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \cdot U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = I_{j,n_1-j+1} \circ F_Y(y)$. Lemma 3.1 implies that

$$\begin{aligned}
h_{1,2,j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(x) \\
&= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \left(F_Y \circ F_Y^{-1}\right) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\
&= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x).
\end{aligned}$$

   The above is the upper-bound of an integral with respect to Xs in the proof of Theorem 3.3. The influence of the distributional specification on Y does not influence the calculation of the paradox probability result in Theorem 3.3.

Proof. First, Lemma 3.1 allows us to write

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{1,2,j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. To see this, consider the random variable $X_{2,(r_{2,k})}$. The random variable $X_{2,(r_{2,k})} = F_X^{-1} \circ U_{(r_{2,k})}$ where

$U_{(r_{2,k})} \sim \text{Beta}(r_{2,k}, N_2 - r_{2,k} + 1)$. Thus, $F_{X_{2,(r_{2,k})}}(x) = g_{N_2,r_{2,k}} \circ F_X(x)$, where we write $g_{n,i}(x) = I_{i,n-i+1}(x)$ for ease of notation. This expression of distribution functions allows us to write

$$
\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,r_{1,j}}^{-1} \circ g_{n_1,j} \circ g_{n_2,k}^{-1} \circ g_{N_2,r_{2,k}} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right] f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,r_{1,j}}^{-1} \circ g_{n_1,j} \circ g_{n_2,k}^{-1} \circ g_{N_2,r_{2,k}} \circ F_X(x)} d\, g_{N_1,r_{1,j}}(F_X(y)) \right] f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,j} \circ g_{n_2,k}^{-1} \circ g_{N_2,r_{2,k}} \circ F_X(x) - g_{N_1,r_{1,j}} \circ F_X(x) \right]^+ f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,j} \circ g_{n_2,k}^{-1} \circ g_{N_2,r_{2,k}} \circ F_X(x) - g_{N_1,r_{1,j}} \circ F_X(x) \right]^+ d\, g_{N_2,r_{2,k}}(F_X(x)) \\
&= \int_0^1 \left[g_{n_1,j} \circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}} \circ g_{N_2,r_{2,k}}^{-1}(t) \right]^+ dt.
\end{aligned}
$$

Condition (4) states that

$$
\int_0^1 \left[g_{n_1,j} \circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}} \circ g_{N_2,r_{2,k}}^{-1}(t) \right]^+ dt = \int_0^1 \left[g_{n_1,j} \circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}} \circ g_{N_2,r_{2,k}}^{-1}(t) \right] dt.
$$

Thus,

$$
\begin{aligned}
&\int_0^1 \left[g_{n_1,j} \circ g_{n_2,k}^{-1}(t) - g_{N_1,r_{1,j}} \circ g_{N_2,r_{2,k}}^{-1}(t) \right]^+ dt \\
&\quad = \int_0^1 g_{n_1,j}(x)\, dg_{n_2,k}(x) - \int_0^1 g_{N_1,r_{1,j}}(x)\, dg_{N_2,r_{2,k}}(x) \\
&\quad = \int_0^1 I_x(j, n_1 - j + 1) \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2 - k + 1)}\, dx \\
&\qquad - \int_0^1 I_x(r_{1,j}, N_1 - r_{1,j} + 1) \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(k, N_2 - k + 1)}\, dx \\
&\quad = \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2 - k + 1)}\, dx \\
&\qquad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{r_{2,k}-1}(1-x)^{N_2-r_{2,k}}}{B(r_{2,k}, N_2 - r_{2,k} + 1)}\, dx \\
&\quad = \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2 + n_1 - p - k + 1)}{B(j, n_2 - k + 1)} \\
&\qquad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2 + N_1 - p - r_{2,k} + 1)}{B(r_{2,k}, N_2 - r_{2,k} + 1)}.
\end{aligned}
$$

□

Often it is of interest to compare the individuals with the highest aptitudes in their respective populations. Below we state the paradox probability (2) in such setting using a Full House Model with two populations with the specification that $F_{Y_1} = F_Y$ for $i = 1, 2$.

Corollary 3.3.1. *Suppose that conditions of Theorem 3.3. Suppose that* $1_1(X_{1,(N_1)}, \boldsymbol{X}_{1,-(N_1)}) = 1_2(X_{2,(N_2)}, \boldsymbol{X}_{2,-(N_2)}) = 1$. *Then,*

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \leq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

A similar result holds for the individuals with the lowest latent aptitudes in the population.

Corollary 3.3.2. *Suppose that conditions of Theorem 3.3. Suppose that* $1_1(X_{1,(1)}, \boldsymbol{X}_{1,-(1)}) = 1_2(X_{2,(1)}, \boldsymbol{X}_{2,-(1)}) = 1$. *Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.$$

Notes:

Proposition 3.4. *Suppose that conditions of Theorem 3.3 where the system inclusion function is* $1_i(X_{i,j} \geq X_{N_i - n_i + 1})$, $i = 1, 2$. *Suppose that* $n_2 \geq n_1 > 1$, $N_1 \geq N_2 > 1$, *and fix* $k = j$. *Then condition (4) holds, and the conclusions of Theorem 3.3 follow.*

Proof. For ease of notation we will let $g_{n,i}(x) = I_{i,n-i+1}(x)$. The proof will use two induction arguments. For the base case let $n_1 = n_2$ and $N_1 = N_2$. With these specifications we have that

$$g_{n_1, n_1 - j} \circ g^{-1}_{n_2, n_2 - j}(x) - g_{N_1, N_1 - j} \circ g^{-1}_{N_2, N_2 - j}(x) = 0.$$

For the induction step, choose some $m \leq n_1 - 1$ and suppose that

$$g_{n_1 - m, n_1 - m - j} \circ g^{-1}_{n_2, n_2 - j}(t) - g_{N_1, N_1 - j} \circ g^{-1}_{N_2, N_2 - j}(t) \geq 0, \tag{5}$$

for $t \in (0, 1)$ where $n_2 \geq n_1$ and $N_1 \geq N_2$. An integration by parts calculation applied to regularized incomplete beta functions gives

$$I_{\alpha,\beta}(x) = I_{\alpha+1,\beta}(x) + \frac{x^\alpha (1-x)^\beta}{\beta B(\alpha, \beta)}.$$

Applying the above to (5) yields

$$\begin{aligned}
&g_{n_1 - m - 1, n_1 - m - 1 - j} \circ g^{-1}_{n_2, n_2 - j}(t) - g_{N_1, N_1 - j} \circ g^{-1}_{N_2, N_2 - j}(t) \\
&\quad = g_{n_1 - m, n_1 - m - j} \circ g^{-1}_{n_2, n_2 - j}(t) - g_{N_1, N_1 - j} \circ g^{-1}_{N_2, N_2 - j}(t) \\
&\qquad + \frac{\left[g^{-1}_{n_2, n_2 - j}(t)\right]^{n_1 - m - 1 - j} \left[1 - g^{-1}_{n_2, n_2 - j}(t)\right]^{j+1}}{(n_1 - m - 1 - j) B(n_1 - m - 1 - j, j + 1)} \\
&\quad \geq 0.
\end{aligned}$$

A similar argument holds for iterative increases to N_1. Thus, condition (4) holds under are assumed conditions and the conclusions of Theorem 3.3 follow. □

Proposition 3.5. *Consider two populations* $i = 1, 2$ *with sizes* N_1 *and* N_2, *and systems of sizes* n_1 *and* n_2. *Suppose that the system inclusion mechanism has the property that* $g(X_{i,j}, \boldsymbol{X}_{i,-j}) = I(X_{i,j} \geq X_{N_i - n_i + 1})$ *for* $i = 1, 2$, *and that the Full House Model is of the form*

$$\begin{aligned}
Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\
X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i,
\end{aligned}$$

where $F_Y^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Suppose that* $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $N_2 - N_1 > n_2 - n_1$, $k = j$. *Then* $g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1, r_{1,k}} \circ g^{-1}_{N_2, r_{2,j}}(t)$ *is non-negative when* $t \in (0, 1)$.

Proof. □

Supposed

$$f(x) = g_{n_1,k} \circ g^{-1}_{n_2,j}(x) - g_{N_1,r_{1,k}} \circ g^{-1}_{N_2,r_{2,j}}(x)$$

Given the property of the system inclusion mechanism, $g(X_{i,j}, \mathbf{X}_{i,-j}) = I(X_{i,j} \geq X_{N_i - n_i + 1})$, and the condtion $k = j$, we have

$$f(x) = g_{n_1,n_1-j} \circ g^{-1}_{n_2,n_2-j}(x) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(x)$$

$$\begin{aligned} f^{'}(x) &= g^{'}_{n_1,n_1-j}(g^{-1}_{n_2,n_2-j}(x)) * \frac{1}{g^{'}_{n_2,n_2-j}(g^{-1}_{n_2,n_2-j}(x))} - g^{'}_{N_1,N_1-j}(g^{-1}_{N_2,N_2-j}(x)) * \frac{1}{g^{'}_{N_2,N_2-j}(g^{-1}_{N_2,N_2-j}(x))} \\ &= \frac{B(n_2 - j, j+1)}{B(n_1 - j, j+1)} g^{-1}_{n_2,n_2-j}(x)^{n_1 - n_2} - \frac{B(N_2 - j, j+1)}{B(N_1 - j, j+1)} g^{-1}_{N_2,N_2-j}(x)^{N_1 - N_2} \end{aligned}$$

When $x = 1$,

$$\begin{aligned} f^{`}(1) &= \frac{B(n_2 - j, j+1)}{B(n_1 - j, j+1)} - \frac{B(N_2 - j, j+1)}{B(N_1 - j, j+1)} \\ &= \frac{(n_2 - j - 1)!(j)!/n_2!}{(n_1 - j - 1)!(j)!/n_1!} - \frac{(N_2 - j - 1)!(j)!/N_2!}{(N_1 - j - 1)!(j)!/N_1!} \\ &= \frac{(n_1 - j)\dots(n_1)}{(n_2 - j)\dots(n_2)} - \frac{(N_1 - j)\dots(N_1)}{(N_2 - j)\dots(N_2)} \end{aligned}$$

Supposed $h(x) = \frac{n_1 - x}{n_2 - x} - \frac{N_1 - x}{N_2 - x}$, then

$$\begin{aligned} h(x) &= \frac{(n_1 - x)(N_2 - x) - (n_2 - x)(N_1 - x)}{(n_2 - x)(N_2 - x)} \\ &= \frac{n_1 N_2 - n_2 N_1 - (n_1 + N_2 - n_2 - N_1)x}{(n_2 - x)(N_2 - x)} \end{aligned}$$

Now we prove for all $0 < x < min(n_1, n_2, N_1, N_2)$, $h(x) < 0$ when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$, $N_2 - N_1 > n_2 - n_1$.

Finally, we show

$$\begin{aligned} f^{`}(1) &= \frac{(n_1 - j)(n_1 - j + 1)\dots(n_1)}{(n_2 - j)(n_2 - j + 1)\dots(n_2)} - \frac{(N_1 - j)(N_1 - j + 1)\dots(N_1)}{(N_2 - j)(N_2 - j + 1)\dots(N_2)} \\ &= \frac{n_1 - j}{n_2 - j} \times \frac{n_1 - j + 1}{n_2 - j + 1} \times \dots \times \frac{n_1}{n_2} - \frac{N_1 - j}{N_2 - j} \times \frac{N_1 - j + 1}{N_2 - j + 1} \times \dots \times \frac{N_1}{N_2} < 0 \end{aligned}$$

We also have $f^{`}(0) = 0$. And we know $f(0) = 0$, and $f(1) = 0$. Therefore, $f(x) \geq 0$ for all $x \in (0, 1)$.

4 Paradox Under Different Distribution

Definition 4.1. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for $\varepsilon > 0$, $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for all y and $\mathrm{P}(Y_2 \leq y) < \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for some y.*

Proposition 4.2. *Suppose that the system inclusion mechanism has the property that $g(X_{1,(r_{1,k})}) = 1$ and $g(X_{2,(r_{2,j})}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_{Y_i}(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_{Y_i}^{-1}(y)$ and $F_X^{-1}(x)$ exist. Further suppose that Y_2 is first order ε-weakly stochastically dominant over Y_1 for some $\varepsilon > 0$. Then,

$$\begin{aligned} \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1 + n_2 - i} - \frac{N_2}{N_1 + N_2} \end{aligned} \tag{6}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \leq \left(F_{Y_1}(y)\right)^{n_1} \leq \left(F_{Y_2}(y) + \varepsilon\right)^{n_1} = \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F_X^{-1}\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2} + \varepsilon\right)^{n_1}\right).$$

In combination with Corollary 3.3.1 and the proof of Corollary 3.3.1, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2}.$$

This verifies the first inequality of (6). We now verify the second inequality of (6). Now,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad \leq \int_{-\infty}^{\infty} \left[\int_x^{F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2} + \varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy\right] f_{X_{2,(N_2)}}(x)dx \\ &\quad = \int_{-\infty}^{\infty} \left[\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} - F_X(x)^{N_1}\right] N_2 F_X(x)^{N_2 - 1} f_X(x)dx \\ &\quad = \int_{-\infty}^{\infty} \left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon\right)^{n_1} N_2 F_X(x)^{N_2 - 1} f_X(x)dx - \frac{N_2}{N_1 + N_2} \end{aligned}$$

$$= \int_0^1 \left(t^{1/n_2} + \varepsilon\right)^{n_1} dt - \frac{N_2}{N_1 + N_2}$$
$$= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1 + n_2 - i} - \frac{N_2}{N_1 + N_2}$$

□

Example: Uniform distributions. In this example we demonstrate a case where Prop 4.2 applies, but we can explicitly calculate the paradox probability. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned} Y_{1,j} &\sim U(0,1), \qquad j = 1, \ldots, n_1, \\ Y_{2,j} &\sim U(0, 1+\delta), \qquad j = 1, \ldots, n_2, \\ X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $\delta > 0$. We can pick $\varepsilon = \delta/(1+\delta)$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Prop 4.2 are satisfied. However, our setup allows for direct calculation of the paradox probability. From Lemma 3.1 and a bit of manipulation, we have that

Theorem 4.3.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}\left(\frac{1}{(1+\delta)}\right)^{n_2} - & \\ \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}, & \text{otherwise} \end{cases}.$$

Proof.

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &= \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (0,1)\right) \\ &+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (1,1+\delta)\right) \\ &= \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < f_{1,2}(X_{2,(N_2)}), F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (0,1)\right) \\ &+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (1,1+\delta)\right) \\ &= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_x^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \\ &+ \int_{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})}^{+\infty} \int_x^{+\infty} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \\ &= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_x^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\, d(F_X(x)^{N_2}) \end{aligned}$$

$$+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}})$$

The support for the outer integration above is $\{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}}) > x\} \cup \{x < F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})\}$.

Therefore, when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{0}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2}) \\
&+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \frac{N_1}{N_1+N_2} - (\frac{1}{1+\delta})^{n_2}\frac{n_1}{n_1+n_2}
\end{aligned}$$

Therefore, when $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$, and let $d = (1+\delta)^{\frac{1}{\frac{N_1}{n_1} - \frac{N_2}{n_2}}}$

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{d^{N_2}}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2})
\end{aligned}$$

$$+ (1 - (\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1 - (\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}})$$
$$= \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}(\frac{1}{(1+\delta)})^{n_2} - \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}$$

□

Here we see that the probability of paradox is 1 for all $\delta \geq [2(n_1/n_2+1)]^{1/n_1} - 1$. When $n_1 \gg n_2$, then $[2(n_1/n_2+1)]^{1/n_1} - 1 \approx n_1^{n_1} - 1$ and we see that the probability of paradox is 1 for extremely small values of δ.

Theorem 4.4. *Assume $g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$ is non-negative when $t \in (0,1)$. Then*

$$\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) \leq (1+\delta)^p \sum_{p=n_1-k}^{n_1} \binom{n_1}{p} \frac{B(n_2-j+p, n_1-p+j+1)}{B(n_2-j, j+1)}$$
$$- \sum_{p=N_1-k}^{N_1} \binom{N_1}{p} \frac{B(N_2-j+p, N_1-p+j+1)}{B(N_2-j, j+1)}$$

Proof.

$$\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) = \mathrm{P}\left(X_{2,(N_2-j)} < X_{1,(N_1-k)} < h_{1,2}\left(X_{2,(N_2-j)}\right)\right)$$
$$= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} f_{X_{1,(N_1-k)}}(y)\, dy \right] f_{X_{2,(N_2-j)}}(x)\, dx$$
$$= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} d\, g_{N_1,k}(F_X(y)) \right] f_{X_{2,(N_2-j)}}(x)\, dx$$
$$= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x) \right]^+ f_{X_{2,(N_2-j)}}(x)\, dx$$
$$= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x) \right]^+ d\, g_{N_2,j}(F_X(x))$$
$$= \int_0^1 [g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)]^+\, dt$$
$$= \int_0^1 g_{n_1,k}((1+\delta)x)\, dg_{n_2,j}(x) - \int_0^1 g_{N_1,k}(x)\, dg_{N_2,j}(x)$$
$$= \int_0^1 I_{(1+\delta)x}(n_1-k, k+1) \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\, dx - \int_0^1 I_x(N_1-k, k+1) \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\, dx$$
$$= \int_0^1 \sum_{p=n_1-k}^{n_1} \binom{n_1}{p} ((1+\delta)x)^p (1-(1+\delta)x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\, dx$$
$$- \int_0^1 \sum_{p=N_1-k}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\, dx$$

$$\leq (1+\delta)^p \int_0^1 \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) x^p (1-x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j, j+1)}\, dx$$

$$- \int_0^1 \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j, j+1)}\, dx$$

$$= (1+\delta)^p \sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array} \right) \frac{B(n_2-j+p, n_1-p+j+1)}{B(n_2-j, j+1)} - \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array} \right) \frac{B(N_2-j+p, N_1-p+j+1)}{B(N_2-j, j+1)}$$

Example: Normal distributions

In this example, we demonstrate a case where Prop 4.2 applies. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned} Y_{1,j} &\sim N(\mu_1, \sigma_1^2), \qquad j = 1, \ldots, n_1, \\ Y_{2,j} &\sim N(\mu_2, \sigma_2^2), \qquad j = 1, \ldots, n_2, \\ X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $\mu_1 \leq \mu_2, \sigma_1 > 0$ and $\sigma_2 > 0$.

Theorem 4.5. *Assume* $\Phi(\frac{\Phi^{-1}(t^{\frac{N_2}{n_2}})\sigma_2 + \mu_2 - \mu_1}{\sigma_1})^{\frac{n_1}{N_1}} > t$ *for all* $t \in (0,1)$,

When $\sigma_1 = \sigma_2 = \sigma$,

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma}, \ldots, \frac{\mu_2 - \mu_1}{\sigma}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}. \end{aligned}$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with* (n_1+n_2-1)*-dimensional mean vector*

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1) \times (n_1+n_2-1)$ *covariance matrix*

$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}$$

The first n_1 *elements of* $\mathbf{w}$ *is* $\frac{\mu_2 - \mu_1}{\sigma}$ *and the last* $n_2 - 1$ *elements are 0.*

When $\sigma_1 \neq \sigma_2$,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}.$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with* $(n_1 + n_2 - 1)$*-dimensional mean vector*

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1 + n_2 - 1) \times (n_1 + n_2 - 1)$ *covariance matrix*

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}_{n_1 \times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1) \times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}_{(n_2-1) \times (n_2-1)}$$

The first n_1 *elements of* $\mathbf{w}$ *is* $\frac{\mu_2 - \mu_1}{\sigma_1}$ *and the last* $n_2 - 1$ *elements are 0.*

Proof. Supposed $\sigma_1 = \sigma_2 = \sigma$, it is obvious to see $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y)$ and we can also find the ε such that $\mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$

$$\begin{aligned} F_{Y_1}(y) - F_{Y_2}(y) &= \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu_1)^2}{2\sigma^2}} dx - \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu_2)^2}{2\sigma^2}} dx \\ &= \int_{y-\frac{\mu_2-\mu_1}{2}}^{y+\frac{\mu_2-\mu_1}{2}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}} dx \end{aligned}$$

Since $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}}$ is the density function for $N(\frac{\mu_2+\mu_1}{2}, \sigma^2)$, then the maximum value of the difference between these two cumulative distribution function of the normal distribution is achieved when $y = \frac{\mu_2+\mu_1}{2}$.

We can pick $\varepsilon = \int_{\mu_1}^{\mu_2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}} dx$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Prop 4.2 are satisfied.

From Lemma 3.1 and a bit of manipulation, we have that

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right)$$

$$
\begin{aligned}
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}})+\frac{\mu_2-\mu_1}{\sigma}))^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} ((\Phi(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{0}^{1} (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt \\
&= \int_{0}^{1} (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1}\ dt - \frac{N_2}{N_1+N_2} \\
&= \int_{-\infty}^{+\infty} \Phi(x+\frac{\mu_2-\mu_1}{\sigma})^{n_1}(n_2\Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1+N_2} \\
&= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1}-X \le \frac{\mu_2-\mu_1}{\sigma}, \ldots, Z_{1,n_1}-X \le \frac{\mu_2-\mu_1}{\sigma}, \\
&Z_{2,1}-X \le 0, \ldots, Z_{2,n_2-1}-X \le 0 | X=x)\phi(x)\ dx - \frac{N_2}{N_1+N_2} \\
&= n_2 \mathbb{P}(Z_{1,1}-X \le \frac{\mu_2-\mu_1}{\sigma}, \ldots, Z_{1,n_1}-X \le \frac{\mu_2-\mu_1}{\sigma}, Z_{2,1}-X \le 0, \ldots, Z_{2,n_2-1}-X \le 0) - \frac{N_2}{N_1+N_2} \\
&= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma}, \ldots, \frac{\mu_2-\mu_1}{\sigma}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}
\end{aligned}
$$

where

$$
\begin{aligned}
h_{1,2}(x) &= (g_1 \circ f_2)(x) \\
&= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\
&= F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}}) - \frac{\mu}{\sigma}))^{\frac{n_1}{N_1}})(x)
\end{aligned}
$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with (n_1+n_2-1)-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1) \times (n_1+n_2-1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma}$ and the last n_2-1 elements are 0.

Then when $\sigma_1 \neq \sigma_2$, we do not have a closed form for ϵ but we can calculate the probability of the paradox.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right)$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}(\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{-\infty}^{+\infty} ((\Phi(\frac{(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx$$

$$= \int_{0}^{1} (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt$$

$$= \int_{0}^{1} (\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}\ dt - \frac{N_2}{N_1+N_2}$$

$$= \int_{-\infty}^{+\infty} \Phi(\frac{\sigma_2 x+\mu_2-\mu_1}{\sigma_1})^{n_1}(n_2\Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \leq \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \leq -\frac{\mu_2-\mu_1}{\sigma_1},$$

$$Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0 | X = x)\phi(x)\ dx - \frac{N_2}{N_1+N_2}$$

$$= n_2 \mathbb{P}(Z_{1,1} - \frac{\sigma_2}{\sigma_1}X \leq \frac{\mu_2-\mu_1}{\sigma_1}, \ldots, Z_{1,n_1} - \frac{\sigma_2}{\sigma_1}X \leq -\frac{\mu_2-\mu_1}{\sigma_1}, Z_{2,1} - X \leq 0, \ldots, Z_{2,n_2-1} - X \leq 0)$$

$$- \frac{N_2}{N_1+N_2}$$

$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}$$

where

$$h_{1,2}(x) = (g_1 \circ f_2)(x)$$

$$= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x)$$

$$= F_X^{-1}((\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{\frac{n_1}{N_1}})(x)$$

$$\Phi^{-1}(F_X(x)^{\frac{N_1}{n_1}})\sigma_1 + \mu_1 < \Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2 + \mu_2$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $(n_1 + n_2 - 1)$-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1 + n_2 - 1) \times (n_1 + n_2 - 1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}_{n_1 \times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1) \times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}_{(n_2-1) \times (n_2-1)}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0. □

5 Examples

5.1 Theory supporting simulations

In this section we verify Theorem 3.3, Theorem 4.3, and 4.5 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 100. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (1) where $X \sim$ Pareto($\alpha = 1.16$). Then $1(X_{(N_1} > X_{N_2}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The results are tabulated in Theorem 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

5.2 Batting averages in Major League Baseball

In this analysis, we compute the probability of the paradox using the results from the Normal distribution example above and illustrate it in the application of Major League

Setting	Distribution for Y_1	Distribution for Y_2	Prob from Theorem	Prob from simulation
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 40$	$N(0,1), n_2 = 60$	0.1495	0.1472
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 30$	$N(0,1), n_2 = 70$	0.2866	0.2875
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 60$	$N(0,1), n_2 = 40$	0	0
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 70$	$N(0,1), n_2 = 30$	0	0
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 40$	$U(0,1+\delta), n_2 = 60$	0.1233	0.1224
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 30$	$U(0,1+\delta), n_2 = 70$	0.2203	0.2210
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 60$	$U(0,1+\delta), n_2 = 40$	0.0022	0.0021
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 70$	$U(0,1+\delta), n_2 = 30$	0.0008	0.0007
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 40$	$N(1,4), n_2 = 60$	0.3338	0.3356
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 30$	$N(1,4), n_2 = 70$	0.3336	0.3343
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 10$	$N(1,4), n_2 = 100$	0.3331	0.3345
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 100$	$N(1,4), n_2 = 10$	0.2378	0.2406
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 40$	$N(1,1), n_2 = 60$	0.3321	0.3326
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 30$	$N(1,1), n_2 = 70$	0.3313	0.3325
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 10$	$N(1,1), n_2 = 100$	0.3334	0.3346
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 100$	$N(1,1), n_2 = 10$	0.2593	0.2592

Table 1: Simulation results to validate the theory

Baseball (MLB) since it is widely acknowledged that the batting average follows Normal distribution.

Specifically, we compute the probability that the top batting average in 2023 is lower than the top batting average in the other season, but the players who have the highest batting average in 2023 are more talented than those who have the highest batting average in the other season.

In this analysis, we detail the considerations made for the batting average statistics that we era-adjust using the Full House Model [Eck, 2020b]. Park factor adjustment is considered in this analysis and we apply the adjusted park index from Schell [2016] to all ballparks from the 1871 season to the 2022 season. The batting average can be affected by the ballpark and we apply the park factor adjustment to this statistic, which accounts for the handedness of batters.

The Full House Model will only be applied to the statistics obtained by full-time players. We define the full-time hitter cutoff as the median PAs after screening out hitters who appeared in fewer than 75 PAs. This criterion is flexible enough to account for changing the number of games played over time as well as seasons shortened by labor strikes and pandemics.

In the 1994 and 1995[1], hitters and pitchers played fewer games per season than the regular season. To deal with extreme statistics from the small sample size, we employed a shrinkage method that adjusted the raw statistics toward a global average. Our shrinkage method follows a shrinkage of ballpark effect estimates motivated from Schell [2016]. The details can be found in Eck [2020b].

We compute the talent scores for the batting average in each season using the (**??**). Then the theoretical results from the Normal distribution example above can be applied to this example and compute the paradox probability of all the seasons from 1871 to 2022 compared with 2023. Fig 1 displays the tendency of the paradox probability of all the seasons from 1871 to 2022 compared with 2023.

[1]https://en.wikipedia.org/wiki/1994%E2%80%9395_Major_League_Baseball_strike

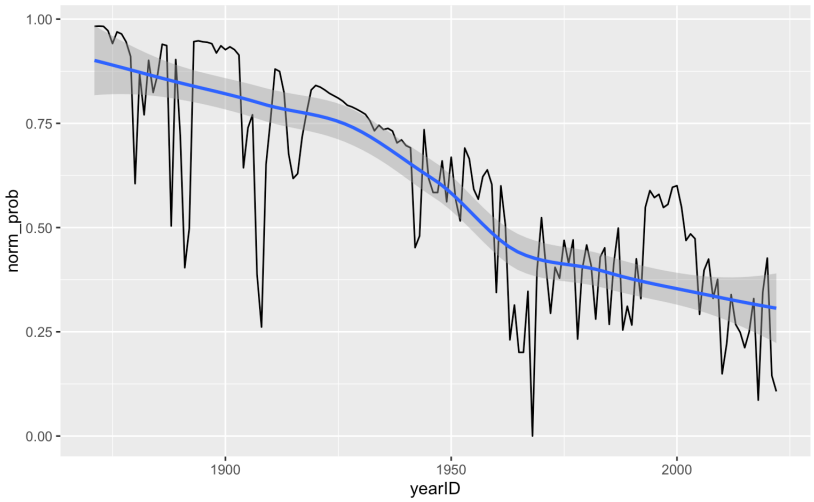


Figure 1: The paradox probability of all the seasons from 1871 to 2022 compared with 2023

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then
$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0,1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha, \beta)$ is the CDF of Beta(α, β) distribution.

References

Björn Andersson, Kenny Bränberg, and Marie Wiberg. Performing the kernel method of test equating with the package kequate. *Journal of Statistical Software*, 55:1–25, 2013.

Scott M Berry, C Shane Reese, and Patrick D Larkey. Bridging different eras in sports. *Journal of the American Statistical Association*, 94(447):661–676, 1999.

John P Campbell. Modeling the performance prediction problem in industrial and organizational psychology. 1990.

Daniel J Eck. Challenging nostalgia and performance metrics in baseball. *Chance*, 33(1): 16–25, 2020a.

Daniel J Eck. Challenging WAR and other statistics as era-adjustment tools. *FanGraphs*, 2020b. https://community.fangraphs.com/challenging-war-and-other-statistics-as-era-adjustment-tools/.

Stephen Jay Gould. *Full House: The Spread of Excellence from Plato to Darwin*. Harvard University Press, 1996.

Kirsten Meyer. Talents, abilities and educational justice. *Educational Philosophy and Theory*, 53(8):799–809, 2021.

Alexander M Petersen and Orion Penner. Renormalizing individual performance metrics for cultural heritage management of sports records. *Chaos, Solitons & Fractals*, 136:109821, 2020.

Alexander M Petersen, Orion Penner, and H Eugene Stanley. Methods for detrending success metrics to account for inflationary and deflationary factors. *The European Physical Journal B*, 79(1):67–78, 2011.

Michael J Schell. *Baseball's all-time best hitters*. Princeton University Press, 2013.

Michael J Schell. *Baseball's all-time best sluggers*. Princeton University Press, 2016.

Amy Zhao Yu, Shahar Ronen, Kevin Hu, Tiffany Lu, and César A Hidalgo. Pantheon 1.0, a manually verified dataset of globally famous biographies. *Scientific data*, 3(1):1–16, 2016.